OIL PRICES OVER TIME AND THEIR IMPACT ON AIRFARES AND BAGGAGE FEES

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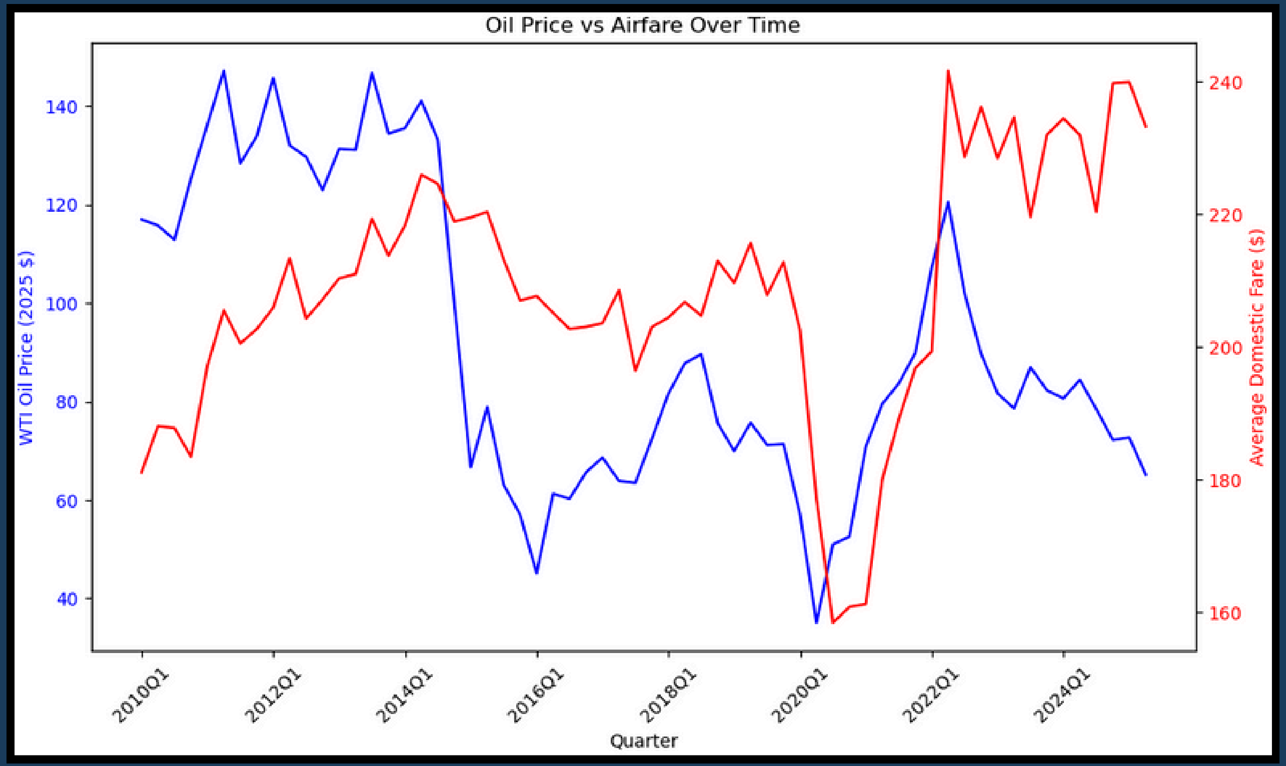
Project Objective

The objective of this project was to find a relationship between oil prices, airline baggage revenues, and airfares. Our initial hypothesis was that there would be a strong correlation between airfares and oil prices and baggage fees and oil prices however from our analysis we found that those correlations are not as strong as assumed. We found that other factors were even better predictors of airfares and baggage fees.

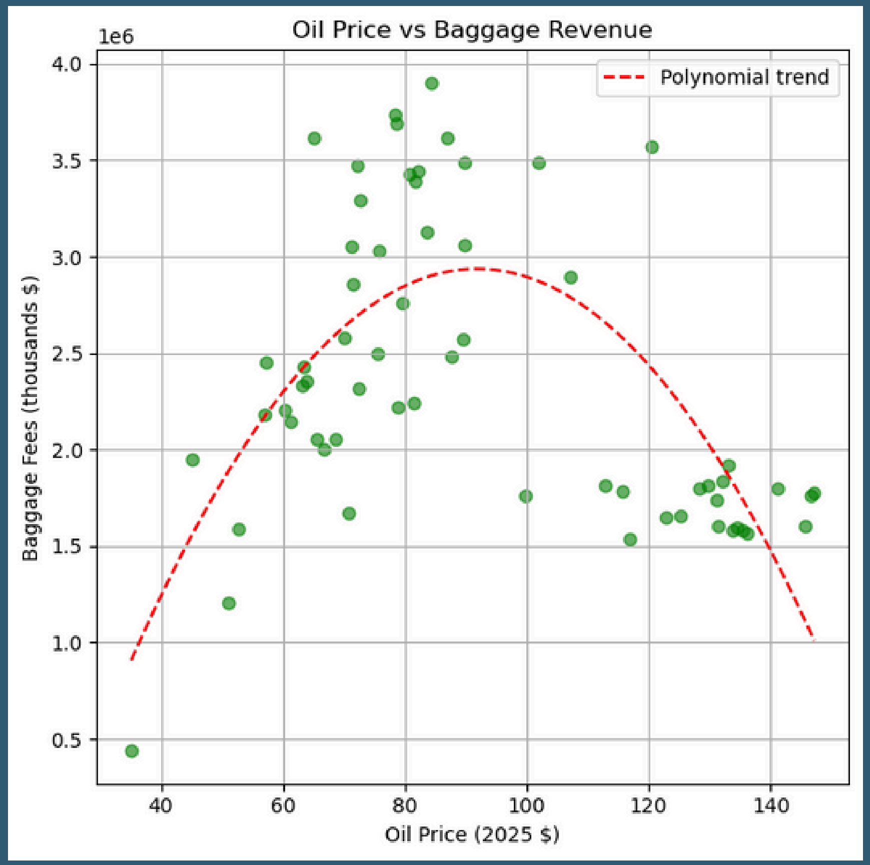
For a more detailed overview watch this video on our project: CS2316 Project Video



Airfares vs Oil Prices

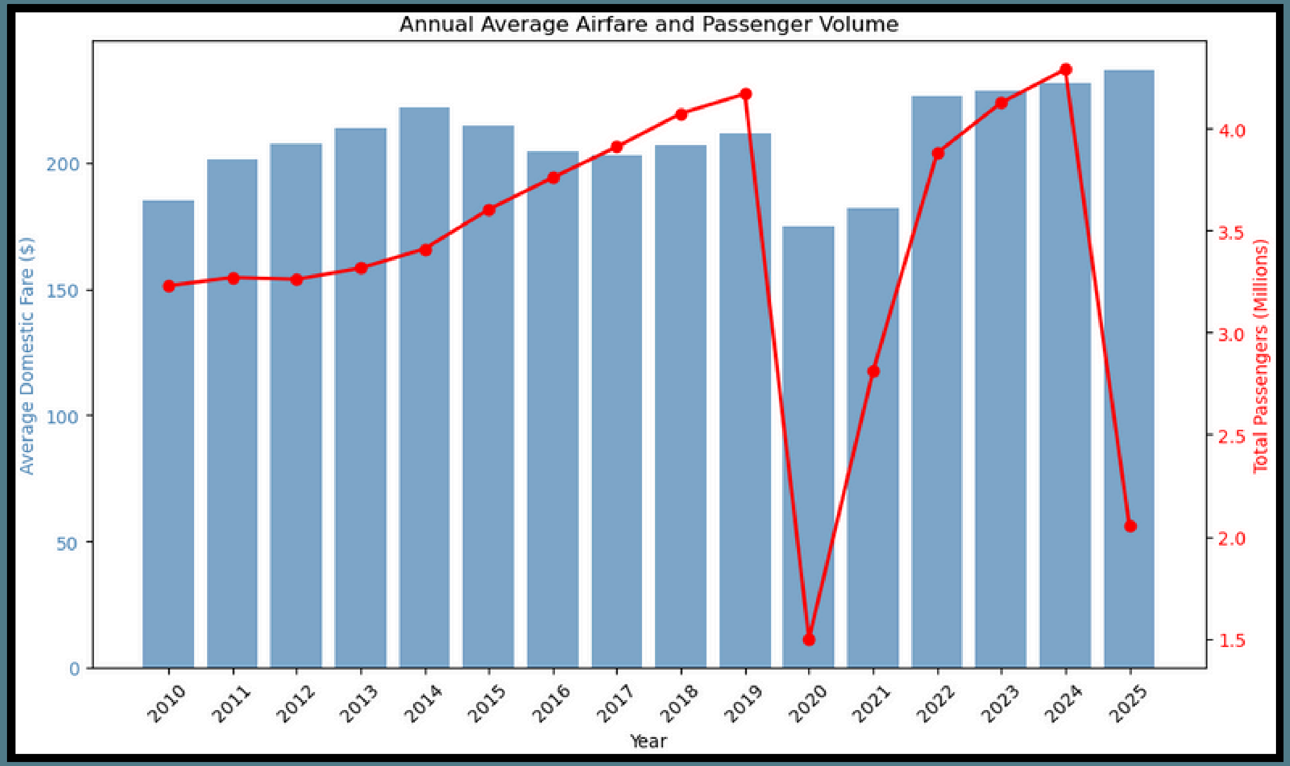
As seen in the chart Oil Prices and Airfares don't seem to have a strong correlation. In fact this correlation was about 0.12 with an R^2 of -0.16 indicating a weak negative correlation. We got this by performing a linear regression.

We then plotted Oil Price and Airfares over time and found that it was a highly variable relationship that didn't line up at all. This indicated that Oil Price simply didn't correlate well with Airfares.



Baggage Revenue vs Oil Prices

We then looked at the relationship between baggage revenues and OIl Prices and it seemed to follow a polynomial trend the best with an R^2 value of 0.51 which indicates a moderate correlation. The baggage fees hit a peak when oil prices are about $90/barrel. Overall, it seems Oil Prices may be able to predict Baggage Revenue to a limited degree



Average Airfares vs Passenger Volume

We then looked at Average Airfares compared to passenger volume since our regression forest found that passenger volume had the highest linear correlation to airfares. Looking at this chart it seems like this is a more accurate predictor of airfare however there is nuance since in 2020 passenger volume dropped sharply but a similar decline in airfares was not seen.